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سورة النجم ٢٩

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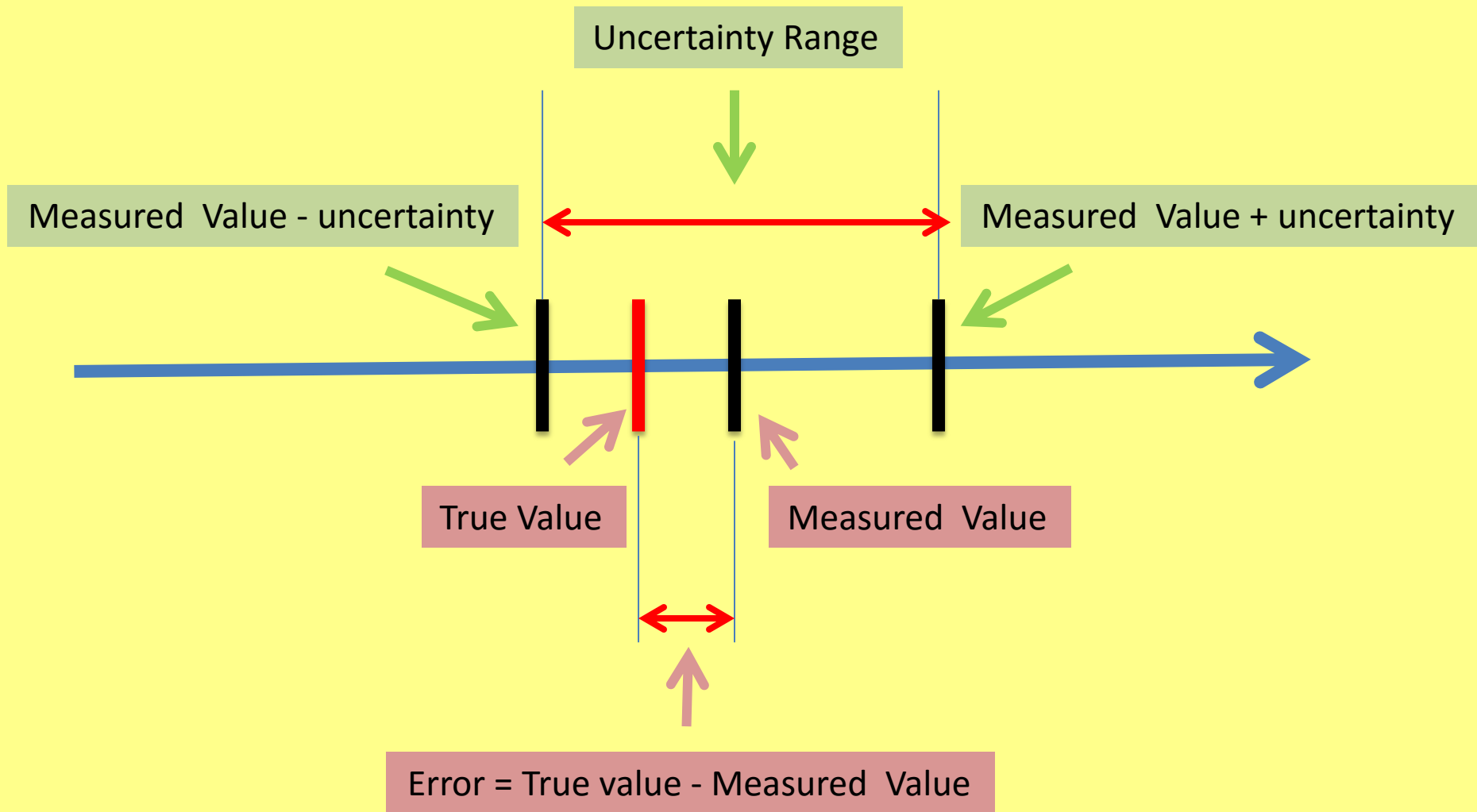
10. References:

1. Measurement

1.1 What is a measurement? a quantity intended to be measure.

- ✓ A measurement tells us about a property of something. It might tell us how heavy an object is, or how hot, or how long it is.
- ✓ A measurement gives a number to that property.
- ✓ Measurements are always made using an instrument of some kind. Rulers, stopwatches, weighing scales, and thermometers are all measuring instruments.
- ✓ The result of a measurement is normally in two parts: a number and a unit of measurement, e.g. 'How long is it? ... 2 meters.'

- ✓ In principle, the aim of a measurement is to obtain the **true value** of the measurand.
- ✓ Every effort is made to optimize the **measurement procedure** in such a way that the measured value is as close as possible to the true value
- ✓ The difference between the measured value and the true value is called **error**. Error can have either positive or negative sign.
- ✓ Error can be regarded as being composed of two parts – **random error** and **systematic error** .



1.2 What is not a measurement?

There are some processes that might seem to be measurements, but are not.

For example : comparing two pieces of string to see which is longer is not really a measurement.

1.3 Mean value

One of the most common approaches for improving the reliability of measurements is making **replicate measurements** of the same quantity.

In such a case very often the measurement result is presented as the **mean value** of the replicate measurements.

In the case of pipetting n times with the same pipette volumes $V_1, V_2, \dots, V_n$ are obtained and the mean value V_m is calculated as follows:

$$V_m = \frac{V_1 + V_2 + \dots + V_n}{n} = \frac{\sum_{i=1}^n V_i}{n}$$

1.4 Standard deviation

The scatter of values obtained from repeated measurements is characterized by **standard deviation** of pipetted volumes, which for the same case of pipetting is calculated as follows:

$$S(V) = \sqrt{\frac{\sum_{i=1}^n (V_i - V_m)^2}{n-1}}$$

1.5 Standard uncertainty

A standard uncertainty is usually shown by the symbol u (small u)

1.6 Coverage factor k

we may wish to have an overall uncertainty stated with **the level of confidence**. This can be done using a coverage factor, k .
as follow : $U = k u_c$

$$k=1 \quad U = \pm u_c : \text{confidence } 68\%$$

$$K= 2 \quad U = \pm 2 u_c : \text{confidence } 95\%$$

**Most
commonly,**

1.7 The relative standard uncertainty

The **relative standard uncertainty** $u_r(y)$ of a measurement result y is defined by

$$u_r(y) = u(y)/|y|,$$

where y is not equal to 0.

1.8 Expanding uncertainty

For those people who use measure and analysis **(CUSTOMERS)** if you use **standard uncertainty** , and informed him that the probability 68% , this means that 32% of the result is out of this range.

For **this reason** the uncertainty result is reported with the expanding uncertainty .

The standard uncertainty denoted by **small u (V)** while the expanding uncertainty denoted by capital U (V)

While **$U(V) = u(V) \cdot K$**
k is the coverage factor

Example : $u_c(V) = 0.019 \text{ ml}$, and $k = 2$; Then $U(V) = 0.019 \times 2 = 0.038 \text{ ml}$

1-9 Standard deviation of the mean

when we use the standard deviation of the **individual value** and
when we use the standard deviation of **the mean**?:

- ✓ whenever the **individual result** is used in further calculation the **standard deviation of the individual** result has to be used
- ✓ whenever the **mean value** is used in further calculations, the **standard deviation of the mean** has to be used.

$$s(V_m) = \frac{s(V)}{\sqrt{n}}$$

Note : When the mean value is calculated from a set of individual values which are randomly distributed then the mean value will also be a random quantity. As for any random quantity, it is also possible to calculate standard deviation for the mean $s(V_m)$. simply divide the $s(V)$ by square root of the number of repeated measurements made:

1.10 The combined standard uncertainties

When estimating the **standard uncertainty** of an output quantity then the standard uncertainties of **all input quantities** are taken into account, the standard uncertainty of the output quantity obtained in this way is called **combined standard uncertainties**, and is denoted by : $u_c(Y)$

1.10.1 Direct measurements

This is the typical way of calculating combined standard uncertainty if all the **uncertainty components** refer to the same quantity and are expressed in the same units.

Example

In case if we need to calculate Combined Standard Uncertainties, from the following individual standard uncertainty

I- repeatability

$$u = (V, \text{Rep})$$

II- calibration:

$$u = (V, \text{Cal})$$

III- Temperature effect:

$$u = (V, \text{Temp})$$

$$u_c(V) = \sqrt{(u(V, \text{Rep.})^2 + u(V, \text{Cal.})^2 + u(V, \text{Temp})^2}$$

1.10.2 Indirect measurements

Indirect measurement is one where the output quantity is found from several “ *not the same quantity as direct measurement*”

In the general case if the output quantity Y is found from input quantities $X_1, X_2, \dots X_n$ according to some function F as follows

$$Y = F (X_1, X_2, X_3, \dots X_n)$$

Then the combined standard uncertainty of the output quantity $u_c(y)$ can be expressed via the standard uncertainties of the input quantities $u(x_i)$ as follows:

$$u_c(Y) = \sqrt{\left(\frac{\partial Y}{\partial x_1} u(x_1)\right)^2 + \left(\frac{\partial Y}{\partial x_2} u(x_2)\right)^2 + \dots + \left(\frac{\partial Y}{\partial x_n} u(x_n)\right)^2}$$

Example

Find the combined standard uncertainty $u_c(y)$ for Density for the following liquid.

$$D = \frac{W}{V}$$

$$u_c(D) = \sqrt{\left(\frac{\partial D}{\partial v} u(v)\right)^2 + \left(\frac{\partial D}{\partial w} u(w)\right)^2}$$

You have the following data :

$w = (11.000 \pm 0.064) \text{ gr} , k = 2, \text{ normal distribution}$

$v = (10.000 \pm 0.042) \text{ ml} , k = 2, \text{ normal distribution}$

$$\frac{\partial D}{\partial v} = -\frac{W}{V^2} = -\frac{11}{100} = -0.11$$

$$\frac{\partial D}{\partial w} = \frac{1}{V} = \frac{1}{10} = 0.1$$

$$u_c(D) = \sqrt{(-0.11 \times 0.021)^2 + (0.1 \times 0.032)^2}$$

$$u_c(D) = \pm 0.012 \text{ gr / ml}$$

1.11 *Pooled standard deviation*

The Pooled Standard Deviation is a weighted average of standard deviations for two or more groups.

It will be explained in more detail with example later

Note : *pooled standard deviation will be explained later in more detail with example (see clause 7)*

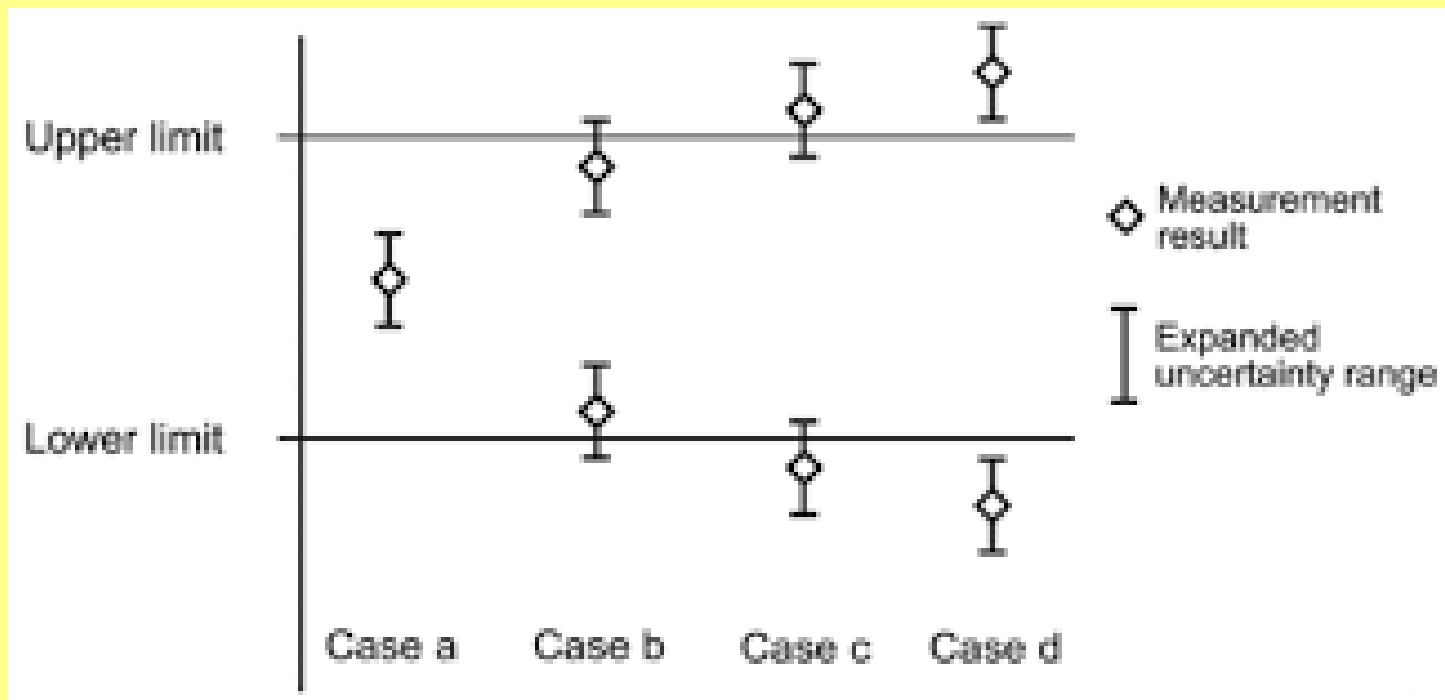
1.12 Compliance with specification

When conclusions are drawn from measurement results, the uncertainty of the measurements *must not be forgotten*.

Sometimes a *result* may fall clearly *inside or outside the limit* of a specification.

But the uncertainty may *overlap the limit*.

Four kinds of outcome are shown in the illustration figure



In Case (a): both the result and the uncertainty fall inside the specified limits. This is classed as a 'compliance'.

In Case (d), neither the result nor any part of the uncertainty band falls within the specified limits. This is classed as a 'non-compliance'.

Cases (b) and (c) are neither completely inside nor outside the limits. No firm conclusion about compliance can be made.

2. Uncertainty of measurement

2.1 What is uncertainty of measurement?

Uncertainty of measurement is the doubt that exists about the result of any measurement.

For every measurement - even the most careful - there is always a margin of doubt.

2.2 Expressing uncertainty of measurement

While there is always a margin of doubt about any measurement, we need to ask 'How big is the margin?' and 'How bad is the doubt?' Thus, two numbers are really needed in order to quantify an uncertainty.

One is the width of the margin, or *interval*. The other is a confidence level.

For example:

We might say that the length of a certain stick measures 20 centimeters plus or minus 1 centimeter, at the 95 percent confidence level

2.3 Error versus uncertainty

It is important not to confuse the terms 'error' and 'uncertainty'.

Error : is the difference between the measured value and the 'true value' of the thing being measured.

Uncertainty : is a quantification of the doubt about the measurement result.

2.4 Why is uncertainty of measurement important?

You may be interested in uncertainty of measurement simply because you wish to make good quality measurements and to understand the results.

However, there are other more particular reasons for thinking about measurement uncertainty.

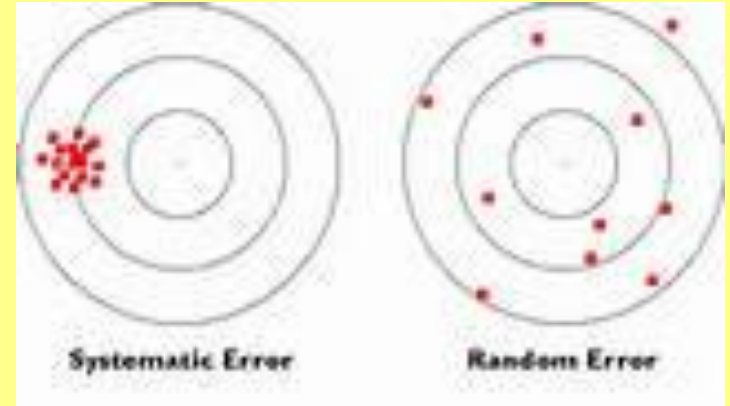
You may be making the measurements as part of a:

- calibration - where the uncertainty of measurement must be reported on the certificate
- test - where the uncertainty of measurement is needed to determine a pass or fail

2.5 The general kinds of uncertainty in any measurement

Random or systematic

The effects that give rise to uncertainty in measurement can be either:



- **Random** : where repeating the measurement gives a randomly different result.
- **systematic** : where the same influence affects the result for each of the repeated measurements.

2.6 What is not a measurement uncertainty?

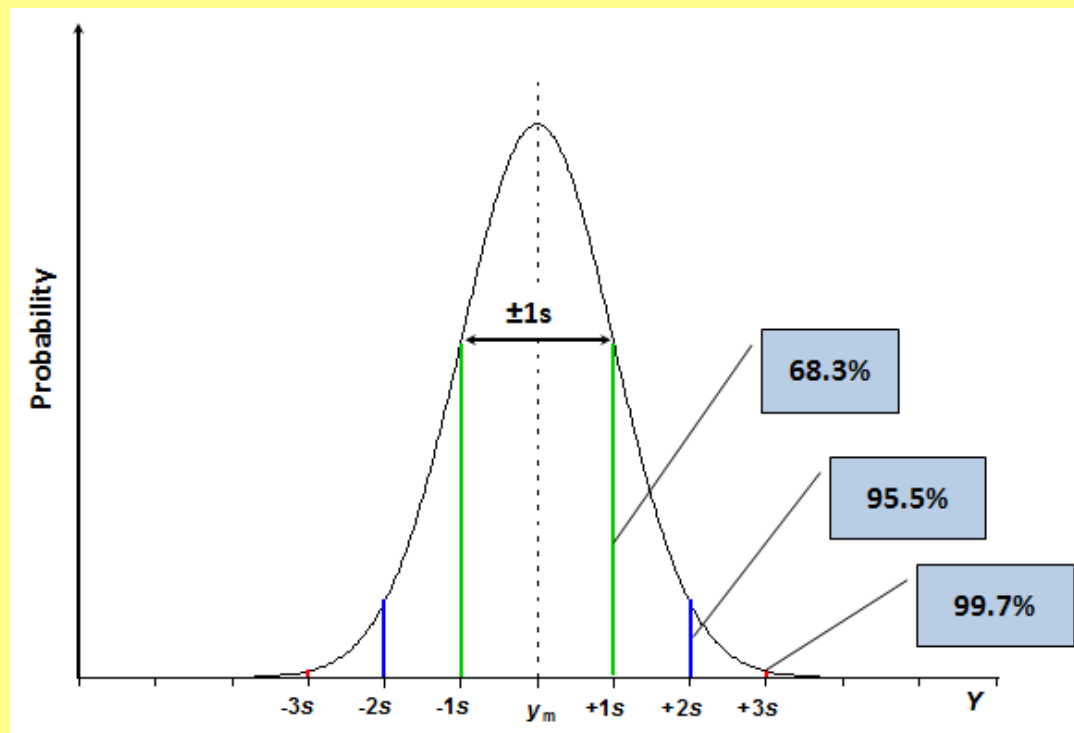
- ✓ Mistakes made by operators are not measurement uncertainties. They should be avoided by working carefully and by checking work.
- ✓ Tolerances are not uncertainties. They are acceptance limits which are chosen for a process or a product.
- ✓ Specifications are not uncertainties. A specification tells you what you can expect from a product.
- ✓ Accuracy is not the same as uncertainty, 'accuracy' is a qualitative term (e.g. you could say that a measurement was 'accurate' or 'not accurate').

Uncertainty is quantitative.

3. Distribution “the shape of the error”

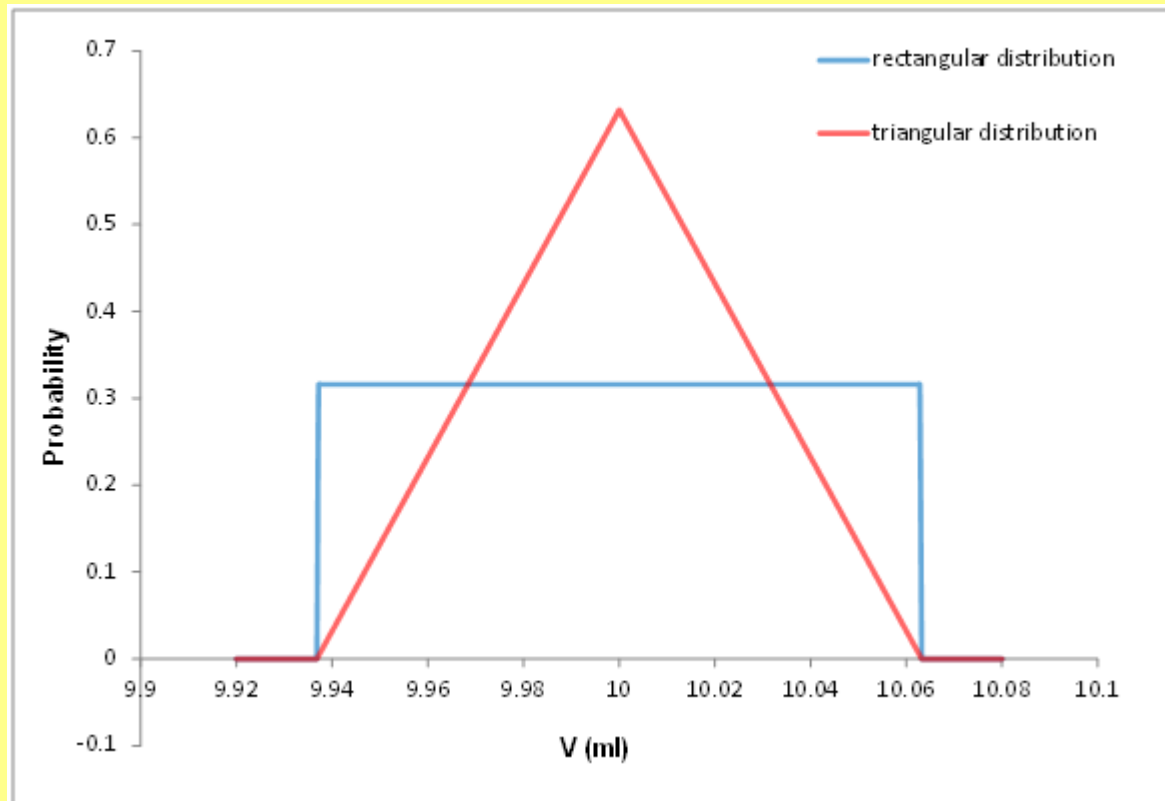
3-1 The normal distribution

The most important distribution in measurement science – the **Normal distribution** (also known as the Gaussian distribution) - the parameters of the Normal distribution (**mean** and **standard deviation**).



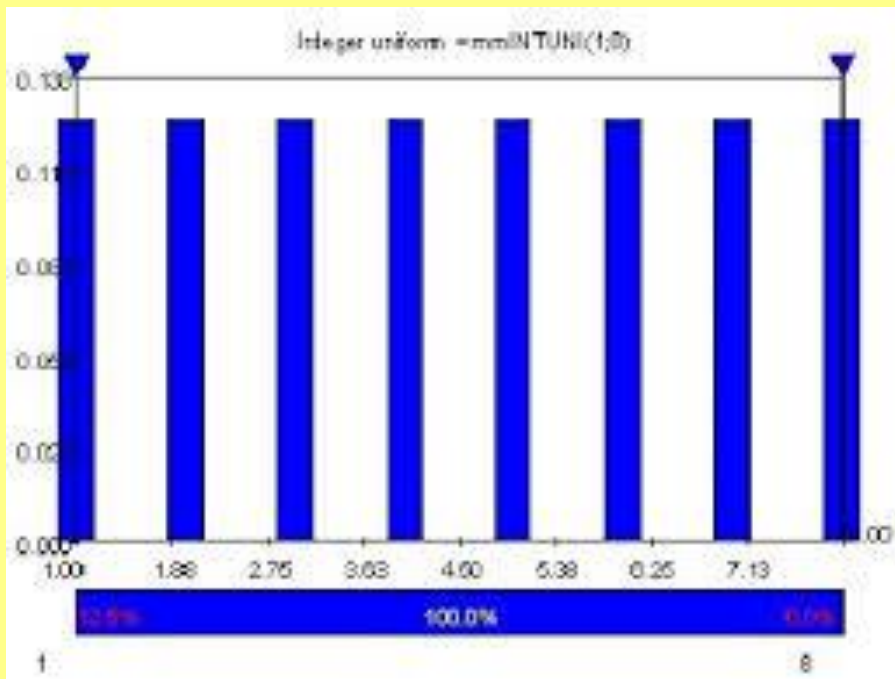
3-2 Rectangular and triangular distribution

- ✓ The uncertainties corresponding to **rectangular or triangular** distribution can be converted to standard uncertainties.
- ✓ Often the **information on distribution function is missing** and then usually some distribution function is assumed or postulated..



3.2.1 Rectangular distribution

When the measurements are quite evenly spread between the highest and the lowest values, a rectangular or uniform distribution is produced.



Example:

If we measure pH, and the value we got = 7.42

In case if we asked for the third decimal point), Then the value may be: 7.415, 7.416, 7.417, 7.418, 7.419, 7.420, 7.421, 7.422, 7.423, 7.424

All at equal probability.

Example 1 :

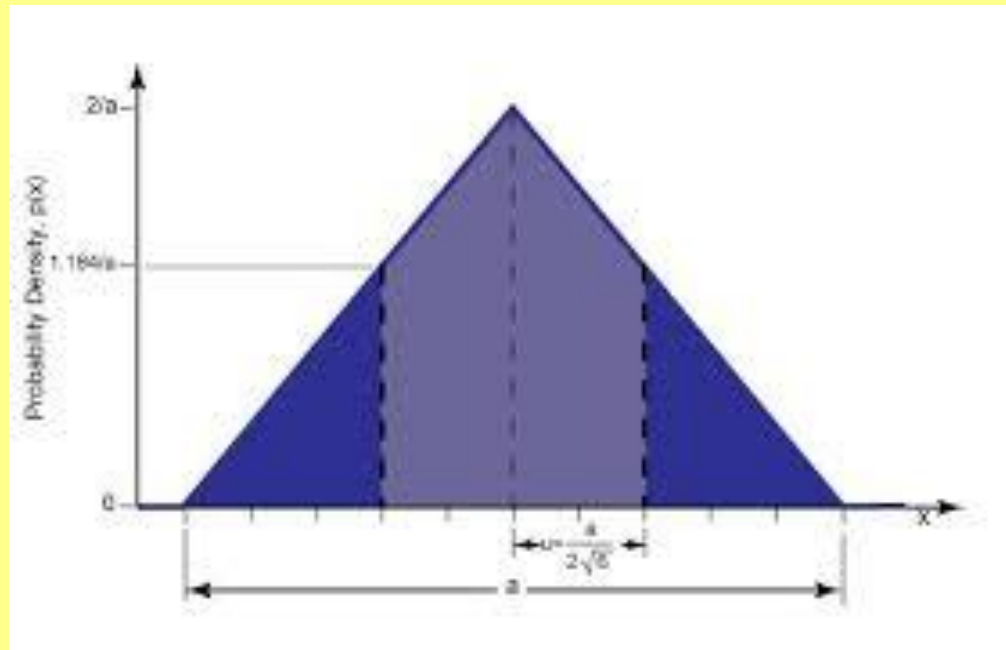
For the shown pipette, the uncertainty for calibration is ± 0.03 ml , and no information to tell us about the distribution. If we assume the rectangular distribution
Then the standard uncertainty :

$$u(\text{cal.}) = \frac{0.03}{\sqrt{3}} = 0.017\text{ml}$$



3.2.2 Triangular distribution :

The triangular is a **continuous, bounded probability distribution** serves as a convenient way to express your uncertainty by providing the **minimum «min»**, and **maximum «max» values** and space of **all possible values** is bounded by «min» and «max».



Example 2 :

For the shown pipette, the uncertainty for calibration is ± 0.03 ml , and no information to tell us about the distribution. If we assume the triangular distribution
Then the standard uncertainty :

$$u \text{ (cal.)} = \frac{0.03}{\sqrt{6}} = 0.012 \text{ ml}$$



Now in example 1 & 2 :

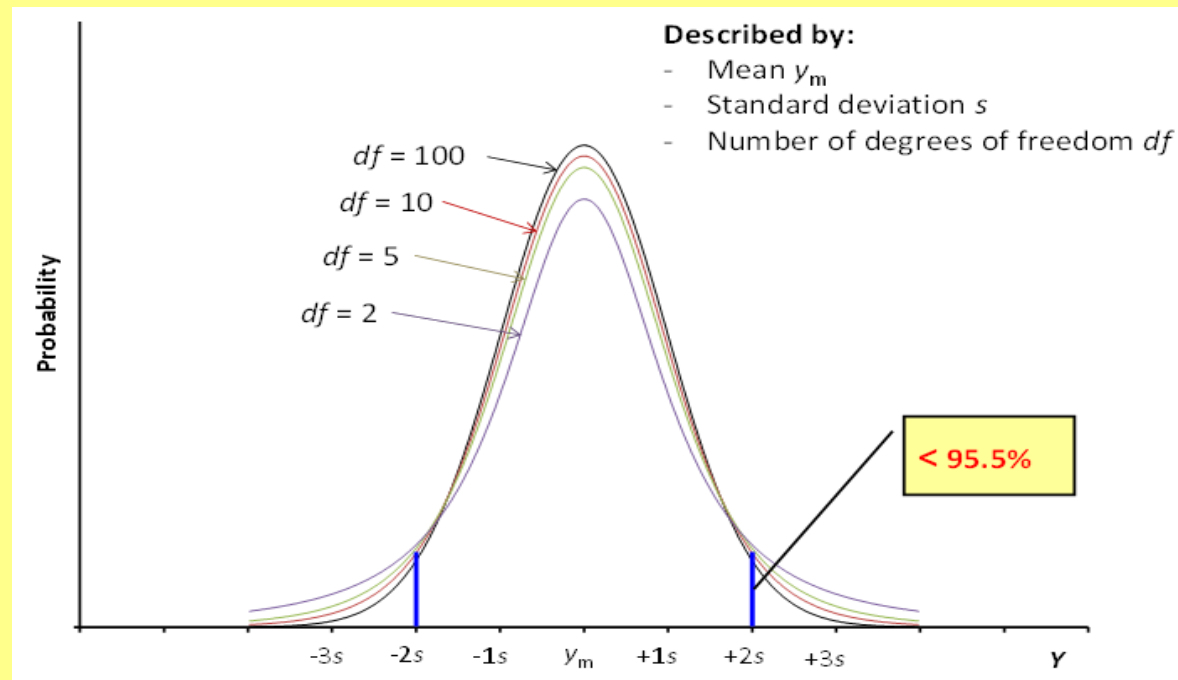
How we chose which of these two distribution (rectangular or triangular), if we don't have information , which one is true?

It is recommended to chose which is under estimated (in our previous 2 examples : is 0.017), then we recommended to select the rectangular distribution .

As general rules, for digital reading , the distribution is rectangular

3-3 The student distribution (also called as *t*-distribution)

Similarly to the normal distribution the Student distribution also has **mean value y_m** and **standard deviation s** . But differently from the normal distribution there is in addition a third characteristic – the number of **degrees of freedom df** .

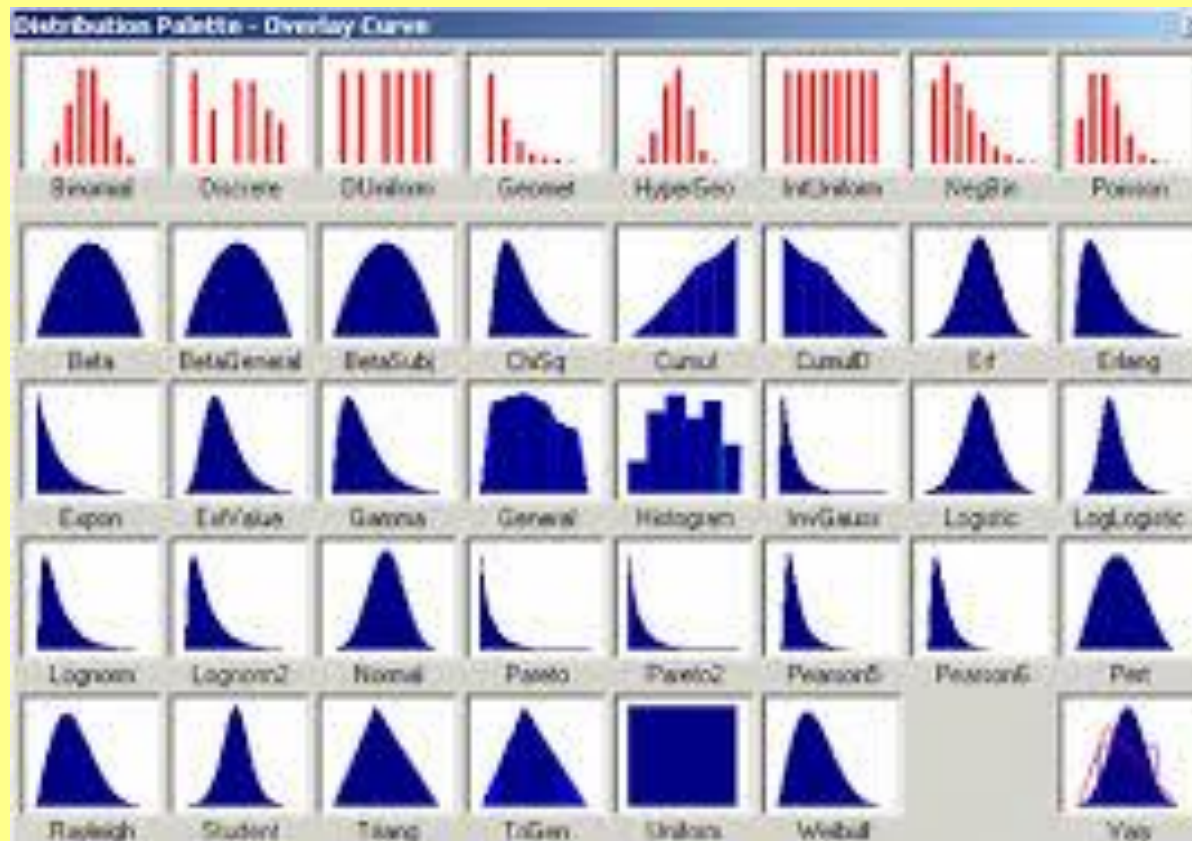


Note : df is equal to the number of repeated measurements minus one.

So, the four Student distribution graphs correspond to 101, 11, 6 and 3 repeated measurements, respectively

3.4 Other distributions

More rarely, distributions can have other shapes, for example, M-shaped (two-peaked), or lop-sided (skew).



4. A and B type uncertainty estimates

A Type : Carrying out the **same measurement (repeated readings)** many times and calculating the **standard deviation** of the obtained values is then the **standard uncertainty** estimate.

B Type : If uncertainty is estimated using some means other than statistical treatment of repeated measurement results then the obtained estimates are called **B type** uncertainty estimates, such as :

- ✓ **Past** experience of the measurements
- ✓ **Calibration** certificates.
- ✓ **Manufacturer**'s specifications
- ✓ **Published** information
- ✓ **Common** sense.

5. Random and Systematic Effects

An effect that within a short time period (e.g. within a day) is systematic can over a longer time period be random. Examples:

Example 1

If a number of pipetting operations are done within a day using the same pipette then the difference of the actual volume of the pipette from its nominal volume (i.e. calibration uncertainty) will be a systematic effect.

If pipetting is done on different days and the same pipette is used then it is also a systematic effect. However, if pipetting is done on different days and different pipettes are used then this effect will change into a random effect.

Example 2

An instrument is calibrated daily with calibration solutions made from the same stock solution, which is remade every month.

In this case the difference of the actual stock solution concentration and its nominal concentration is a systematic effect within a day and also within few weeks.

But over a longer time period, say, several months (at least 4-5) is minimum. This effect becomes random, since a number of different stock solutions will have been in use during that time period.

Conclusions :

- ✓ An effect, which is systematic in short term can be random in long term;
- ✓ The longer is the time frame the more effects can change from systematic into random.

Important notes :

- 1- If the measurement of the same sample is repeated (usually within the same day) using the same procedure then the standard deviation is called **repeatability standard deviation (s_r)**.
- 2- If the measurement of the same sample is repeated using the same procedure but under **changed conditions** then the standard deviation of the results is called **within-lab reproducibility (s_{RW})**.
- 3- Then ; why s_r is smaller than s_{RW} ?. Simply, some effects that **within day are systematic** and **are not accounted** for by s_r ; **become random** over a **longer** time and s_{RW} **takes them into account**.

4- **The combined standard uncertainty u_c** , is in turn **larger**, because it has to take into account all significant effects that influence the result, including those that **remain systematic** also in the long term.

$$\begin{array}{ccccc} \text{Repeatability} & < & \text{Within-lab reproducibility} & < & \text{Combined uncertainty} \\ s_r & < & s_{Rw} & < & u_c \end{array}$$

A & B type uncertainty estimate relation to random and systematic effects

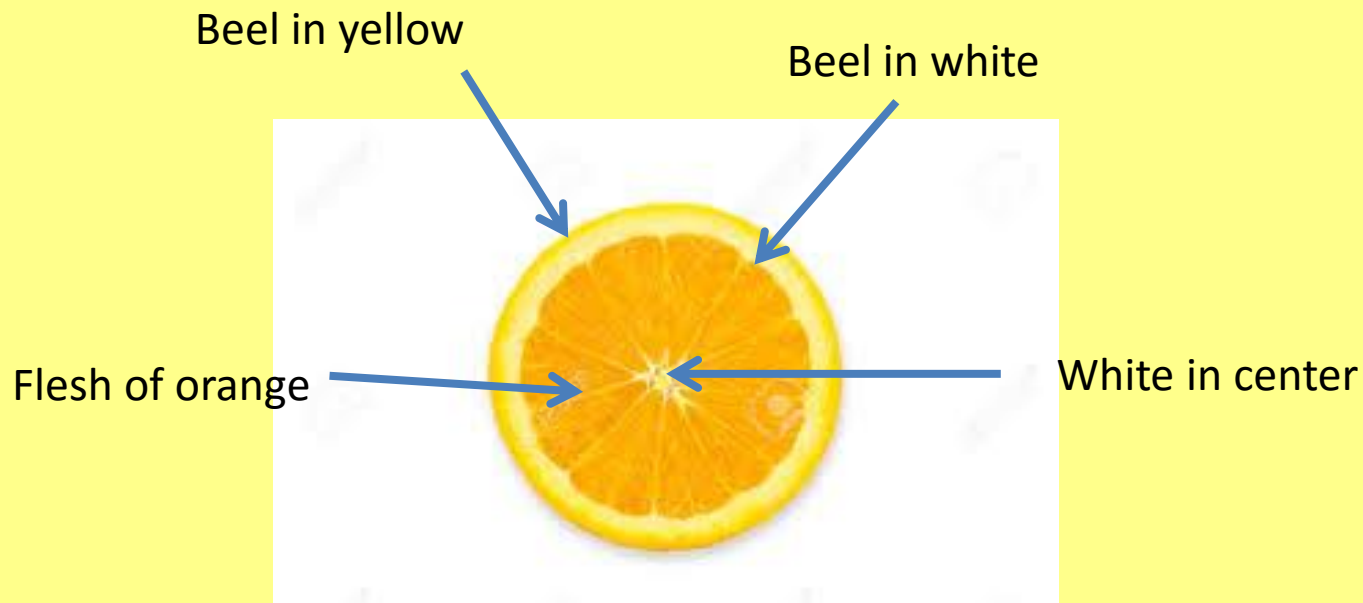
Effect	Type A	Type B
Random	The usual way	Possible if the data on the magnitude of the effect are available without performing repeated measurements
systematic	Only possible if the effect will change into a random effect in the long term	The usual way

6. Principles of Measurement Uncertainty Estimation

6.1 Measurand definition

The first principle of measurement uncertainty is: the **measurand must be correctly defined**.

The importance of measurand definition is explained on the example of **pesticide** determination in **oranges**.



Defining **the measurand** in the case of pesticide determination in oranges **is not trivial**, for :

- ✓ On one hand it is important to define whether the result is applied to a **single orange** or **few oranges** that were taken as the **sample** or whether it is applied to the **whole lot** of oranges (the whole analysis object, also called sampling target).
- ✓ On the other hand, oranges are **not homogenous**. Pesticides are applied on **orange surface**, not inside, at the **same time pesticide can diffuse** from the orange peel to the inside.
- ✓ So, a number of different possibilities exist: whole orange, whole peel, outside part of the peel, only orange flesh.

6.2 Measurement procedure

The main steps of a measurement procedure are presented on the example of pesticide measurement are presented:

- ✓ Sample preparation (in this case: **homogenization, extraction(s), extract purification**).
- ✓ The majority of analytical procedures need that the sample is **converted into a solution** which contains as large as possible share of the analyte from the sample (ideally all of it)
- ✓ Essentially destroying the sample matrix so that a solution containing the **analyte(s) and few matrix components** is obtained. This is often done by **digestion** with acids or fusing with alkalies or salts.
- ✓ Sample preparation is often the **most work-intensive** part of chemical analysis and in most cases it is also the part, which has the **largest uncertainty contribution**.

6.3 Sources of measurement uncertainty

Recommendations

- ✓ Most of the uncertainty sources are **linked to specific steps** in the analysis procedure.
- ✓ It is stressed that **sample preparation** is usually the **biggest** contributor to measurement uncertainty.
- ✓ When performing chemical analysis, care should be taken to **minimize the influence of the uncertainty sources**, as far as possible.
- ✓ And what **cannot be eliminated**, has to be taken into account.
- ✓ It is not necessary to **quantify every uncertainty** source individually.
- ✓ Instead, it is often more practical to **quantify several uncertainty sources jointly**.

Example of Sources of Uncertainty

- ✓ Sampling (**non – representativeness**)
- ✓ Sample **preparation** :
 - Reduction (decomposition)
 - Homogenization (volatilization)
 - Extraction (incomplete extraction)
 - Purification (loss during purification)
 - Concentration (contamination)
- ✓ Instrument analysis (instrument sources such as noise, drift , ...)
- ✓ Calibration with analyte (concentration / purity of calibrants material)

Uncertainty sources

– Sampling

- Sample non-representativeness

– Sample preparation

- Inhomogeneity
- Separation of analyte incomplete
- Analyte adsorbs

- Analyte or photometric complex decomposes
- Analyte volatilizes
- Incomplete reaction
- Contamination

– Preparation and dilution of solutions

– Weighing

– Calibration of instrument

- Standard substance purity
- Solution preparation

– Measurement of sample

- Interferences
- Repeatability of reading
- Drift of reading
- Memory effects

Example 1

How to calculate the standard uncertainty for volumetric pipette



The uncertainty sources effect the pipette

(Repeatability & Calibration & Temperature)

Data given :

- 1- $V = 10 \text{ ml}$**
- 2- standard deviation (repeatability) = 0.006 ml**
- 3- Pipette uncertainty written on the pipette itself = 0.03 ml**
- 4- max exceed temp 24°**
- 5- Factor to change the rectangular shape to normal shape = $\sqrt{3}$**
- 6- γ = Thermal expansion coefficient of water = 0.00021**

Calculation

1- repeatability

$$u = (V, \text{Rep}) = \text{standard deviation} = 0.006 \text{ ml}$$

2- calibration:

$$u = (V, \text{Cal}) = 0.03 / \sqrt{3} = 0.017 \text{ ml}$$

Pipette uncertainty written
on the pipette itself

Factor to change the rectangular
shape to normal shape

3- Temperature effect:

$$u(V, \text{Temp}) = V \cdot \Delta t \cdot \gamma / \sqrt{3} = 0.0049 = 0.005 \text{ ml}$$

$$\Delta t = 24 - 20 = 4^\circ$$

Factor to change the rectangular
shape to normal shape

Combined standard uncertainty (u_c)

The simplest form

$$u_c(V) = \sqrt{(u(V, \text{Rep.})^2 + u(V, \text{Cal.})^2 + u(V, \text{Temp})^2)}$$

$$u_c(V) = \sqrt{0.006^2 + 0.017^2 + 0.005^2} = 0.019 \text{ ml}$$

$$u_c(V) = 0.019 \text{ ml (at } k = 1 \text{)}$$

$$u_c(V) = 0.019 \times 2 = 0.038 \text{ ml (at } k = 2 \text{)}$$

Note : that final standard uncertainty 0.019 **is to close to cal.** standard uncertainty, while the other 2 values **are smaller and have no influence** on the final uncertainty.

Presenting Measurement Results

The correct presentation of measurement result includes **value**, **uncertainty** and information about the **probability** of the uncertainty.

We can assume that $k = 2$ corresponds to roughly 95% of coverage probability.

Then the correct presentation of the measurement result in this case would look as follows:

The volume of the pipetted liquid is :

$$V = (10.000 \pm 0.038) \text{ ml}, k = 2, \text{ normal distribution}$$

Important note 1 :

In this example : there are three main sources of uncertainty: repeatability, calibration and the temperature effect. These effects influence pipetting in different ways.

- 1- Repeatability is a typical **random** effect.
- 2- The uncertainty due to calibration of the pipette is a typical **systematic** effect
- 3- The temperature effect can be, depending on the situation, either **systematic or random** effect or (very commonly) mixture of the two.

Although the **three uncertainty sources** influence pipetting results in **different ways** they are all **taken into account the same way** – via uncertainty contributions expressed as standard uncertainties.

Important note 2 :

if one wants to **reduce** the uncertainty. it is always necessary to focus on **decreasing** the uncertainties caused by the **largest components**.

So, in this example it is not very useful to buy a more expensive air conditioner to reduce “ **the Temperature effect** ”.

It will also **not be possible** to improve the uncertainty by reducing the repeatability component.

Clearly, whatever is done with these two components the combined standard uncertainty $u_c(V)$ cannot decrease from 0.019 ml to lower than 0.017 ml, which is not a significant decrease, **then the best way to reduce the uncertainty is to calibrate the pipette in the laboratory**; see example 2

Example 2

This is an example of calculating the volume and its uncertainty of liquid delivered from a self-calibrated volumetric pipette

The uncertainty of the pipetted volume $u(V)$ has three main uncertainty components (see example 1):

- ✓ uncertainty due to repeatability, $u(V, \text{rep})$;
- ✓ uncertainty due to pipette calibration, $u(V, \text{cal})$
- ✓ uncertainty due to the temperature difference from 20 °C, $u(V, \text{temp})$

In this example we are going to weight the water and calculate the volume (volume = weight / density of pure water)

Important note 3 :

- ✓ Usually in the case of **high-accuracy** work, the pipette is calibrated in the laboratory in order to obtain **lower calibration uncertainty**.
- ✓ if the uncertainty due to factory calibration is used (**example 1**), then this calibration uncertainty component is the most influential one. **So, reducing it would also reduce the overall uncertainty.**
- ✓ It is very important, that calibration is performed under the **same conditions** and preferably **by the same person**.
- ✓ From the calibration data we can obtain two important pieces of information:
 - the correction term for the pipette **volume $V_{\text{correction}}$** with uncertainty **$u(V, \text{cal})$**
 - repeatability of pipetting **$u(V, \text{rep})$** .

Practical data

Temperature of water = 24 °C

Density of pure water ($d = 0.9973 \text{ g / cm}^3$)

Trial no	m (water) g By experimental weighted	V (water) ml By calculation (m/d)
1	9.96037	9.98734
2	9.96454	9.99152
3	9.9632	9.99017
4	9.97152	9.99852
5	9.9683	9.99529
6	9.97413	10.00113
7	9.96806	9.99505
8	9.96165	9.98862
9	9.96397	9.99095
10	9.9544	9.98135

The mean value (average) $V_m = \frac{V_1 + V_2 + V_3 + \dots + V_{10}}{10}$

$$V_m = \frac{9.98734 + 9.99152 + 9.99017 + \dots + 9.98135}{10}$$

$$V_m = 9.992$$

$$\text{Correction} = 9.9920 - 10.0000 = -0.0080$$

$$\text{The standard deviation}(s) = \sqrt{\frac{\sum_i^n (V_i - V_m)^2}{n-1}}$$

$$\text{The standard deviation}(s) = \sqrt{\frac{(V_1 - V_m)^2 + (V_2 - V_m)^2 + \dots + (V_{10} - V_m)^2}{n-1}}$$

The standard deviation(s) =

$$\sqrt{\frac{(9.98734 - 9.992)^2 + (9.99152 - 9.992)^2 + \dots + (9.98135 - 9.992)^2}{10-1}} = 0.0057 \text{ ml}$$

Standard uncertainty due to repeatability $u(V, \text{Rep})$ is equal to this standard deviation = 0.0057 ml.

Standard uncertainty of the calibration is expressed as the standard deviation of the mean; see (clause 1.9)

$$u(V, \text{Cal}) = s(V_m) = \frac{s(V)}{\sqrt{n}} = \frac{0.0057}{\sqrt{10}} = 0.0018 \text{ ml}$$

Uncertainty due to temperature effect:

When there is a possibility that pipetting is performed at a **different temperature** from the **calibration** then an additional uncertainty source due to **temperature change** is introduced and it has to be **taken into account**.

We assume that :

- ✓ pipette's using temperature does not differ from the calibration temperature **by more or less than 4 °C** ($\Delta t = \pm 4^\circ\text{C}$)
- ✓ assuming **rectangular** distribution
- ✓ water's density depends on temperature, therefore we have to consider the thermal expansion coefficient of water, which is $\gamma_w = 2.1 \cdot 10^{-4} \text{ 1/}^\circ\text{C}$.
- ✓ $V = 10.0000 - \text{correction} = 10.0000 - 0.0080 = 9.9920$

$$u(V, \text{Temp}) = V \cdot \Delta t \cdot \gamma / \sqrt{3}$$

$$u(V, \text{Temp}) = \frac{9.9920 \times 4 \times 2.1 \times 10^{-4}}{\sqrt{3}} = 0.0048 \text{ ml}$$

Combined standard uncertainty (u_c)

$$u_c(V) = \sqrt{(u(V, \text{Rep.})^2 + u(V, \text{Cal.})^2 + u(V, \text{Temp})^2)}$$

$$u_c(V) = \sqrt{0.0057^2 + 0.0018^2 + 0.0048^2} = 0.0077 \text{ ml}$$

The $k = 2$ expanded uncertainty of the pipetted volume can be found as follows:

$$U_c(V) = u_c(V) \cdot k = 0.0077 \cdot 2 = 0.0154 \text{ ml}$$

The volume of the pipetted liquid is :

$$V = (9.992 \pm 0.015) \text{ ml}, k = 2, \text{ normal distribution}$$

Compare between example 1 & 2

Item description	Example 1	Example 2
Procedure	Measure volume	Measure weight and calculate volume
$u(V, \text{Rep})$	0.006 ml	0.0057 ml
$u(V, \text{Cal})$	0.017 ml	0.0018 ml
$u(V, \text{Temp})$	0.005 ml	0.0048 ml
$u_c(V)$	0.019 ml	0.0077 ml
Correction	0	0.008 ml
Volume at $k=2$, normal dis.	10.000 ± 0.038	9.992 ± 0.015

7. Pooled Standard Deviation

The Pooled Standard Deviation is a weighted average of standard deviations for two or more groups. See clause (1.11).

If it is impossible to make many repeated measurement with the same sample

Then precision can be estimated during longer time in the form of pooled standard deviation

Pooled standard deviation can be used to calculate

- ✓ Repeatability
- ✓ Within – lab reproducibility

Repeatability measures the variation in measurements taken by :

- ✓ Single instrument
- ✓ The same person
- ✓ Under the same conditions.

Reproducibility : the experiments were conducted numerous times to test the reproducibility of the results

General formula for the case when experiment is done with different samples.

$$S_{\text{pooled}} = \sqrt{\frac{(n_1-1).s_1^2 + (n_2-1).s_2^2 + \dots + (n_k-1).s_k^2}{n_1 + n_2 + \dots + n_k - k}}$$

Symbols:

K = number of samples

$s_1, s_2, \dots, s_k$ are within sample standard deviation

$n_1, n_2, \dots, n_k$ are number of measurement made from different samples

Pooled Standard Deviation gives us the average situation of the lab

Example 3

- ✓ How to set up experiment for repeatability S_r using S_{pooled}

Day	4 /5/2020	5 /5/2020	6 /5/2020	7 /5/2020
sample	1	3	2	4
value	10.59964 11.52049 10.8061 11.04245	13.1775 13.52363 14.35686	12.76157 11.98296 11.71126 11.92767 11.91558	10.645 11.034
S	0.394971	0.606213	0.405625	0.271065
n-1	3	2	4	1
(n-1).S ²	0.468007	0.734989	0.658128	0.075661

Samples

4

3

2

1

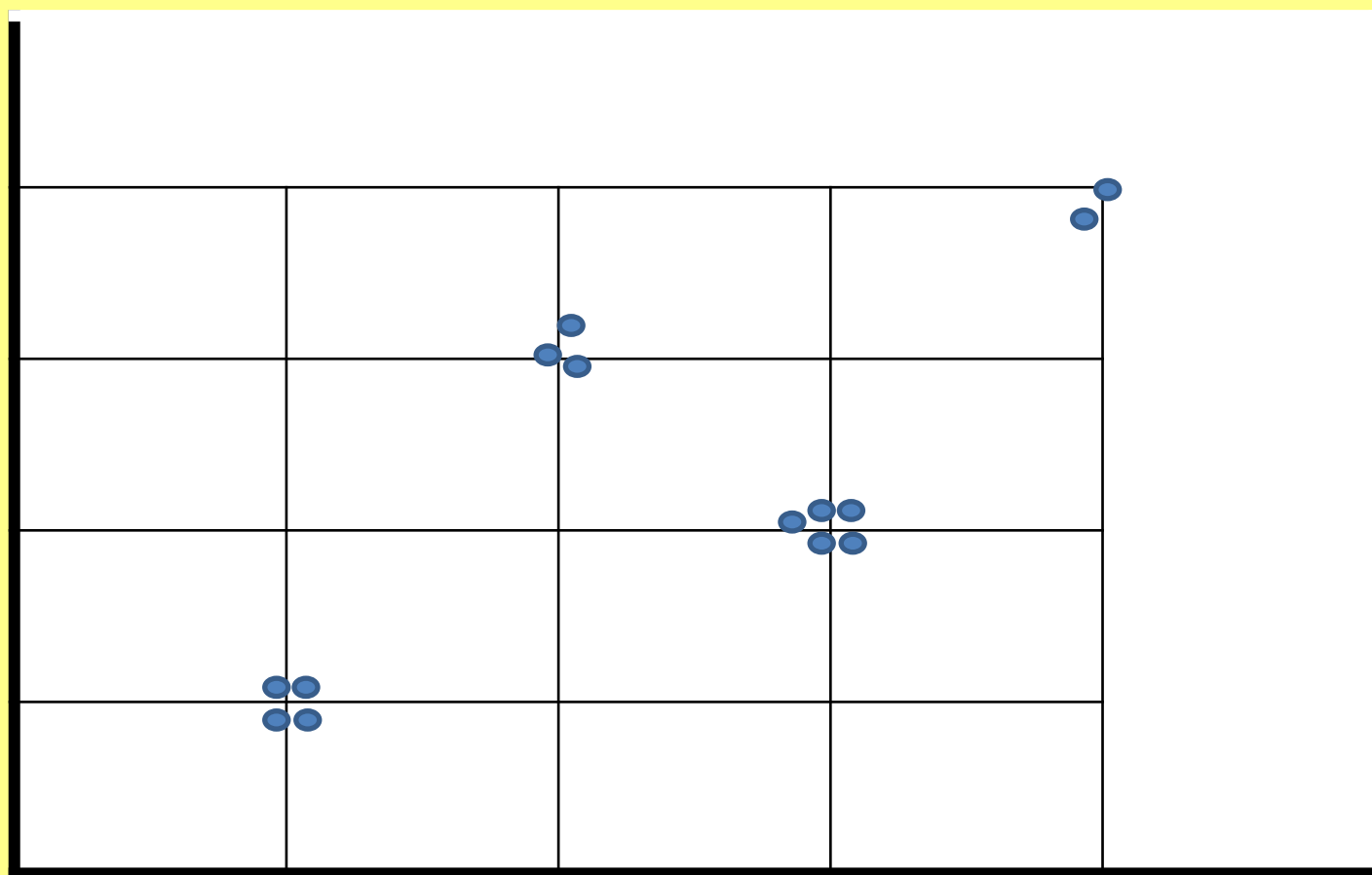
4/5/2020

5/5/2020

6/5/2020

7/5/2020

Days



$$S_{\text{pooles}} = \sqrt{\frac{(n_1-1).s_1^2 + (n_2-1).s_2^2 + \dots + (n_k-1).s_k^2}{n_1 + n_2 + \dots + n_k - k}}$$

$$S_{\text{pooles}} = \sqrt{\frac{0.468007 + 0.734989 + 0.658128 + 0.075661}{4+3+5+2-4}}$$

$$S_{\text{pooles}} = 0.440089 \mu\text{g /g}$$

Example 4

- ✓ How to set up experiment for within long term reproducibility
 S_{RW} using S_{pooled}

Date	Sample no.	Result (g/100g)	Stand. Dev. (g/100g)	n-1	(n-1).S ²
15/1/2020 19/2/2020 19/8/2020	1	10.2 11.2 11.4	0.64291	2	0.82667
19/2/2020 25/3/2020 8/4/2020	2	14.2 13.9 13.5	0.351188	2	0.246667
15/1/2020 19/2/2020 25/3/2020	3	17.6 16.9 17.3	0.351188	2	0.246667
25/3/2020 8/4/2020 10/6/2020	4	12.1 12.7 12.3	0.3055303	2	0.186662
8/4/2020 10/6/2020	5	19.2 18.6	0.424264	1	0.18

Samples

5

4

3

2

1

15/1/2020

19/2/2020

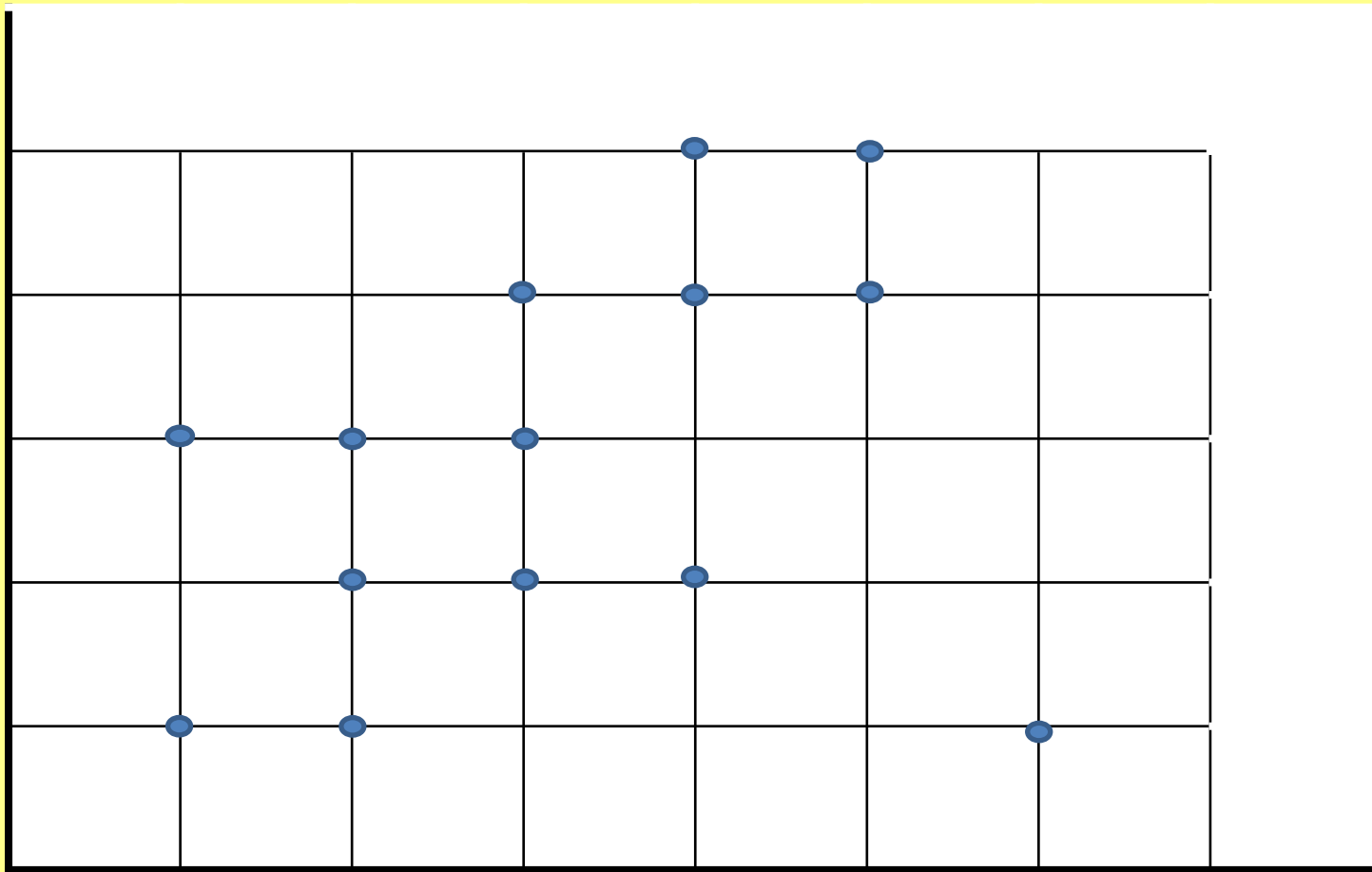
25/3/2020

8/4/2020

10/6/2020

19/8/2020

Days



$$S_{\text{pooles}} = \sqrt{\frac{(n_1-1).s_1^2 + (n_2-1).s_2^2 + \dots + (n_k-1).s_k^2}{n_1+n_2+\dots+n_k-k}}$$

$$S_{\text{pooles}} = \sqrt{\frac{0.82667 + 0.246667 + 0.246667 + 0.186662 + 0.18}{3+3+3+3+2-5}}$$

$$S_{\text{pooles}} = 0.432906 \text{ g /100 g}$$

8. Single-lab Validation Approach

(also called “Nordtest approach”- and also called “top-down” approach)

- ✓ The single-lab validation approach:
 - does not go deeply into the measurement procedure
 - does not attempt to quantify all uncertainty sources individually.
- ✓ Most of the data that are used come from validation of the analytical procedure(this is the reason for the word “validation” in the name of the approach)

Important note 4 :

- ✓ Clauses 8.1 to 8.3 present the Nordtest approach and explain the way of obtaining necessary data.
- ✓ Clause 8.4 gives a “roadmap” of the Nordtest approach.
- ✓ Practical example of applying the Nordtest approach

8.1 Principle

In **Single-lab Validation** (Nordtest) approach the uncertainty is regarded as being due to two components:

- 1- The **within-lab reproducibility component “ $u(R_w)$ ”**; this uncertainty component takes into account all uncertainty sources that are **random in the long term** (some uncertainty sources that are systematic within a day will become random in the long term).
- 2- The **bias component “ $u(bias)$ ”**, this component takes into account the systematic effects that cause long-term bias.

The main equation of the Nordtest approach is :

$$u_c = \sqrt{u(R_w)^2 + u(bias)^2}$$

The main steps of the process of measurement uncertainty evaluation with the **Nordtest approach**:

- ✓ Specify measurand
- ✓ Quantify R_w component $u(R_w)$
- ✓ Quantify bias component $u(bias)$
- ✓ Convert components to standard uncertainties $u(x)$
- ✓ Calculate combined standard uncertainty u_c
- ✓ Calculate expanded uncertainty U

8.2 Uncertainty component accounting for Random Effect

The within-lab reproducibility “ $u(R_w)$ ” takes into account all uncertainty sources that are random in the long term (i.e. several months, preferably one year). So, quite some uncertainty sources that are systematic within a day will become random in the long term.

This is one of the key points of the Nordest approach of uncertainty estimation.

This is also an important advantage of the approach, because estimating random effects can be done more reliably than estimating systematic effects.

Important note 5 :

- ✓ The pooled standard deviation approach can be used. If this is done then the $u(R_w)$ can be found based on several different control samples, so that it will be an average value of all of them.
- ✓ The time period during which the data are collected, must be sufficiently long (at least several months, preferably around a year) so that all the sources of variability in the procedure are taken into account.
- ✓ 10 values collected over a five-month time period is a better option than 20 values collected during one- month

8.3 Uncertainty component accounting for Systematic Effects

Bias is always determined against a reference value, which also has an uncertainty.

- ✓ Bias refers to difference between our measured value and a reference value.
- ✓ Therefore, for finding u (*bias*) we need a sample or a material with a reference value.
- ✓ There are four different possibilities how the u (*bias*) can be determined:
 - From repeated analysis of the same sample with reference procedure.
 - From repeated analysis of certified reference material (CRMs).
 - From repeated interlaboratory comparison measurements.
 - From repeated spiking experiments.

The **u (bias)** component is found according to the following equation:

$$u(\text{bias}) = \sqrt{(RMS_{\text{bias}})^2 + u(C_{\text{ref}})^2}$$

RMS_{bias} is the average (root mean square) bias and is found as follows:

$$RMS_{\text{bias}} = \sqrt{\frac{\sum_i^n (\text{bias}_i)^2}{n}}$$

Where :

- ✓ n is the number of bias determinations carried out .
- ✓ bias_i is a result of an individual bias determination and is found as follows:

$$\text{bias}_i = C_{\text{lab}_i} - C_{\text{ref}_i}$$

- ✓ $u(C_{\text{ref}})$ is the average standard uncertainty of the reference values of the reference samples and is found as follows:

$$u(C_{\text{ref}}) = \sqrt{\frac{\sum_i^n u(C_{\text{ref}_i})^2}{n}}$$

8.4 Road map

$$u_c = \sqrt{u(R_w)^2 + u(\text{bias})^2}$$

```
graph TD; A["u_c = sqrt(u(R_w)^2 + u(bias)^2)"] --> B["u(bias) = sqrt(RMS_bias^2 + u(C_ref)^2)"]; A --> C["u(R_w) = S_Rw"]; B --> D["RMS_bias = sqrt(sum_i^n (bias_i)^2 / n)"]; B --> E["u(C_ref) = sqrt(sum_i^n u(C_refi)^2 / n)"]; D --> F["bias_i = C(lab_i) - C(ref_i)"]; E --> G["u(C_refi) = s_i / sqrt(n_i)"];
```

$$u(\text{bias}) = \sqrt{(\text{RMS}_{\text{bias}})^2 + u(C_{\text{ref}})^2}$$

$$\text{RMS}_{\text{bias}} = \sqrt{\frac{\sum_i^n (\text{bias}_i)^2}{n}}$$

$$u(C_{\text{ref}}) = \sqrt{\frac{\sum_i^n u(C_{\text{refi}})^2}{n}}$$

$$u(R_w) = S_{Rw}$$

$$\text{bias}_i = C(\text{lab}_i) - C(\text{ref}_i)$$

$$u(C_{\text{refi}}) = \frac{s_i}{\sqrt{n_i}}$$

Important note 6 :

Steps of the process :

- ✓ Specify measurand
- ✓ Quantify R_w component .. $u(R_w)$
- ✓ Quantify bias component .. $u(\text{bias})$
- ✓ Convert components to standard uncertainty.. $u(x)$
- ✓ Calculate combined standard uncertainty.. u_c
- ✓ Calculate expanded uncertainty.. $U(x)$

Example 5

Determination of acrylamide in snacks

Example data :

- 1- Concentration level 998 $\mu\text{g/ Kg}$
- 2- Laboratory has analyzed two certified reference materials with similar matrix
 - ✓ Crisp bread & potato chips
 - ✓ The crisp bread is also used as a control sample
- 3- The crisp bread CRM has the following acrylamide content
 - ✓ $C_{\text{acrylamide}} = 1179 \pm 68 \mu\text{g/ Kg}$ (k=2, norm)
- 4- The potato chips CRM has the following acrylamide content
 - ✓ $C_{\text{acrylamide}} = 860 \pm 42 \mu\text{g/ Kg}$ (k=2, norm)

Experimental data

Crisp bread

Days	µg/ Kg
5/1/2018	1172
6/3/2018	1186
3/4/2018	1153
8/1/2019	1151
18/3/2019	1181
3/4/2019	1147
11/4/2019	1097
16/4/2019	1102
25/4/2019	1162
3/8/2019	1138
28/9/2019	1122
27/11/2019	1191

Mean = 1150 µg/ Kg, standard dev. = 31 µg/ Kg

Potato chips

Days	µg/ Kg
3/4/2018	845
3/4/2018	832
3/4/2018	802
27/4/2018	829
27/4/2018	851
27/4/2018	834

Mean = 832 µg/ Kg, standard dev. = 17 µg/ Kg

Mean value and standard dev. Can be calculated from these two equations

$$V_m = \frac{V_1 + V_2 + \dots + V_n}{n} = \frac{\sum_{i=1}^n V_i}{n}$$

$$S(V) = \sqrt{\frac{\sum_{i=1}^n (V_i - V_m)^2}{n-1}}$$

Calculation

1- For random effect , within – laboratory reproducibility calculation

Mean = 1150 µg/ Kg, standard dev. = 31 µg/ Kg (see experimental data - Crisp bread)

$$u (R_w) = S_{Rw} = 31 \mu\text{g/ Kg}$$

$$u (R_w)_{\text{rel.}} = \frac{S_{Rw}}{\text{Mean}} = \frac{31}{1150} = 2.7\%$$

2- For bias calculation

Ref. value for crisp bread (see example data – no. 3)

$$C_{\text{acrylamide}} = 1179 \pm 68 \mu\text{g/ Kg (k=2, norm)}$$

Ref. value for the potato chips (see example data – no. 4)

$$C_{\text{acrylamide}} = 860 \pm 42 \mu\text{g/ Kg (k=2, norm)}$$

u (k= 1) “standard uncertainty”

✓ Crisp bread

$$C_{\text{acrylamide}} = 1179 \pm 34 \mu\text{g/ Kg (k=1, norm)}$$

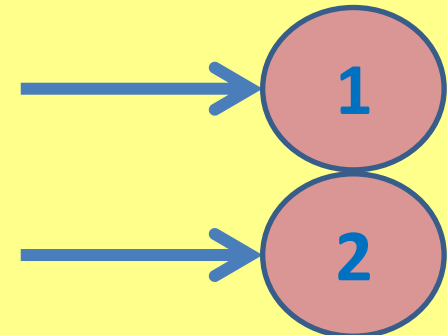
✓ Potato chips

$$C_{\text{acrylamide}} = 860 \pm 21 \mu\text{g/ Kg (k=1, norm)}$$

For u_{rel}

✓ Crisp bread $\underline{u c_{\text{rel}}} = \left(\frac{34}{1179} \times 100 \right) = 2.88\%$

✓ Potato chips $\underline{u c_{\text{rel}}} = \left(\frac{21}{860} \times 100 \right) = 2.44\%$



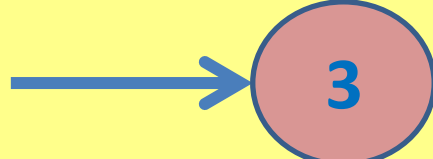
For bias = Laboratory result – CRM Ref value

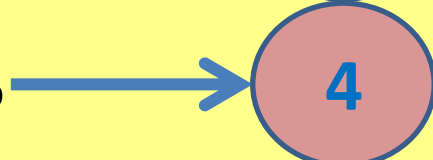
$$\text{bias} = C_{\text{lab}} - C_{\text{ref}}$$

Crisp bread mean value = 1150 µg/ Kg (see experimental data)
Ref. value = 1179 µg/ Kg (see example data – no. 3)
bias = 1150 – 1179 = -29 µg/ Kg

Potato chips mean value = 832 µg/ Kg (see experimental data)
Ref. value = 860 µg/ Kg (see example data – no. 4)
bias = 832 – 860 = -28 µg/ Kg

Bias_{rel.} = bias value / ref. value

Crisp bread = $\frac{-29}{1179} \times 100 = -2.45 \%$ 

Potato chips = $\frac{-28}{860} \times 100 = -3.24 \%$ 

From 1 & 2

$$u(c_{\text{rel}}) = \sqrt{\frac{\sum (c_{\text{rel}_i})^2}{n}}$$

✓ Crisp bread $u c_{\text{rel}} = 2.88\%$

✓ Potato chips $u c_{\text{rel}} = 2.44\%$

$$u(c_{\text{rel}}) = \sqrt{\frac{(2.88)^2 + (2.44)^2}{n=2}} = 2.67\%$$

5

From 3 & 4

$$\text{RMS}_{\text{bias.}} = \sqrt{\frac{\sum (\text{bias})^2}{n}} = \sqrt{\frac{(-2.45)^2 + (-3.24)^2}{2}} = 2.87\%$$

6

From 5 & 6

$$u_c(\text{bias})_{\text{ref}} = \sqrt{(\text{RMS}_{\text{bias}})^2 + u(C_{\text{ref}})^2}$$

$$u_c(\text{bias})_{\text{ref}} = \sqrt{(2.87)^2 + (2.67)^2} = 3.92\%$$

7

3- Combined standard equation

$$u_c = \sqrt{u(R_w)^2 + u(\text{bias})^2}$$

$$u(R_w)_{\text{rel.}} = \frac{S_{RW}}{\text{Mean}} = \frac{31}{1150} = 2.7\% \text{ "Crisp bread" (see experimental data)}$$

From 7 & 8

$$u_c = \sqrt{(2.7)^2 + (3.9)^2} = 4.8\% (k=1)$$

$$U_c = \frac{4.8}{100} \times \text{measurement result value (998 } \mu\text{g/ Kg)} = 48 \mu\text{g/ Kg (k=1)} \\ = 96 \mu\text{g/ Kg (k=2)}$$

Acrylamide content in the sample = $998 \pm 96 \mu\text{g/ Kg (k=2, norm)}$

9. Solved Exam

(100 points)

1- Which of the listed differences between uncertainty and error are correct? (10 points)

Choose :

- a-Error has a sign but uncertainty does not
- b-Uncertainty and error have different units
- c-Uncertainty is surely bigger than the absolute value of error
- d-Uncertainty cannot be used for correcting the measured value, while error in principle could be used (if we knew its direction and magnitude)

Answer ()

2- Which of the following statements about measurement uncertainty, random effects and systematic effects are correct?
(10 points)

Choose :

- a - Correctly estimated measurement uncertainty takes into account both random and systematic effects.
- b-Systematic effects can in principle be corrected and thereby minimized or even eliminated.
- c- Uncertainty sources are the cause of deviation of the measured value from the true value.
- d-It is often possible in practice to eliminate all uncertainty sources so that measurement results will have no uncertainty.
- e- Random effects can never be entirely eliminated but their influence on the result can be decreased by making multiple repeated measurements.
- f-Systematic effects can be minimized (or even eliminated) by performing the measurement in large number of repetitions

Answer ()

3- Which coverage factor has to be used in order to have expanded uncertainty estimate with confidence level of approximately 95%? It is known, that there is normal distribution. (5 points)

Choose one

- a- 3
- b-2.5
- c-2

Answer ()

4- Standard deviation characterizes the scatter of the values obtained from repeated measurements. (5 points)

Choose one

- a-True
- b- False

Answer ()

5- A number of repeated measurements of the same measurand are performed and the mean value and standard deviation are calculated. Which of the following statements are true? (10 points)

Choose

- a - The standard deviation can be calculated according to equation
$$X_m = (X_1 + X_2 + \dots + X_n) / n.$$
- b- The mean is always exactly in the middle of the largest and the smallest obtained value.
- c- In the case x is normally distributed with mean x_m and standard deviation s the probability of a next measurement result of x falling into $x_m \pm s$ is roughly 95%.
- d- Standard deviation expresses the difference of our measurement result from the true value.
- e-Standard deviation can be regarded as an estimate of measurement error.

- f-The mean is always somewhere between the largest and the smallest obtained value.
- g-Standard deviation of a quantity x characterizes the spread of the values obtained from repeated measurements.
- h-The mean can be calculated by summing up all results and dividing the sum with the number of obtained values.
- i-According to properties of the normal distribution the mean value is the most probable value for next result when doing a large number of repetitions
- j-The mean can be calculated according to the equation $X_m = X_1/n + X_2/n + \dots + X_n/n$
- k-The mean can be calculated according to the equation $X_m = (X_1 + X_2 + \dots + X_n) / n$.

Answer ()

6- Calculate the titration repeatability standard deviation when the titrant volumes consumed for titration of identical samples were (all in ml): 12.02; 11.96; 11.99; 12.00; 12.02. (10 points)

Please present your answer with three decimals after comma.

Answer ()

7- Which of the obtained uncertainty estimates is safer (i.e. lower danger to underestimate the uncertainty)? (5 points)

Choose one

a-rectangular distribution

b-triangular distribution

Answer ()

8- Which is more accurate titration procedure, gravimetric or volumetric titration? (5 points)

Choose one

- a- Volumetric
- b- Gravimetric

Answer ()

9- The result of cadmium determination in waste water was 2.3 mg/l with combined standard uncertainty 0.21 mg/l. Please express the expanded uncertainty with 95% coverage probability (normal distribution can be assumed). (5 points)

Answer ()

10- Which of the following is true? There are 5 correct answers. (10 points)

Choose

- a - Standard uncertainty is uncertainty expressed at roughly 68% confidence level
- b- Expanded uncertainty is uncertainty that has been found by taking into account more uncertainty sources than in the case of standard uncertainty
- c- Expanded uncertainty is roughly equal to the measurement bias
- d- Combined standard uncertainty is standard uncertainty that has been calculated taking into account all relevant uncertainty sources
- e-Expanded uncertainty is found by multiplying combined standard uncertainty with a coverage factor
- f-Combined standard uncertainty can be calculated from expanded uncertainty by applying square root

- g - Standard uncertainty can be found, depending on situation, as standard deviation of parallel measurement results or standard deviation of the mean of parallel measurement results
- h- Expanded uncertainty is found by adding the standard uncertainties of all input quantities
- i-Expanded uncertainty expresses the range of values within which the true value lies with 100% probability
- j - Expanded uncertainty is roughly equal to the measurement error
- k -Combined standard uncertainty is standard uncertainty expressed at roughly 95% confidence level
- l-Combined standard uncertainty can be calculated from expanded uncertainty by dividing it with the coverage factor

Answer ()

11- Which are correctly expressed measurement results? (10 points)

Choose :

- a- The measurement result of zinc content in potato sample is in range of 3.4...3.7 mg/kg with probability of approximately 68%.
- b- $c(\text{Pb}) = (0.080 \pm 0.0614) \text{ mg/kg}$, $k=2$, norm.
- c- Manganese content in the water sample is $(3.43 \pm 0.43) \text{ mg/kg}$, $k=3$, norm.
- d- $c(\text{Zn}) = (2.7 \pm 0.30) \%$, $k=2$, norm
- e- $c(\text{Fe}) = (5.85 \pm 0.1) \text{ mg/kg}$, $k=2$, norm.
- f- Iron content in the sample is 2.17 mg/kg with standard uncertainty of 0.05 mg/kg, $k=2$, norm
- g- Dissolved oxygen level in the water is 5.34 mg/l with combined standard uncertainty of 0.17 mg/l.

Answer ()

12 - Which of the following are random effects and which are systematic effects? (10 points)

- a- Titration repeatability ()
- b - Analyte loss during the sample preparation ()
- c- Titrant decomposition during the analysis ()
- d- Matrix effect in liquid chromatography mass spectrometry leading to partial suppression of ionization of the analyte ()
- e- Less than 100% purity of the standard substance used for constructing calibration graph. ()
- f- Instrument reading repeatability ()
- g- Contamination of sample with the analyte ()
- h- Increased absorbance readings in spectrophotometric analysis caused by interfering substance ()

Answer : Random effects ()

Systematic effects ()

13- Accuracy refers to difference between the measured value and true value. (5 points).

Choose one

a- True

b- False

Answer ()

Answer Sheet

Question no	Points	Correct answer	Result
1	10	a & d	
2	10	a & b & c & e	
3	5	c	
4	5	a	
5	10	f & g & h & i & j & k	
6	10	0.025 ml	
7	5	a	
8	5	b	
9	5	0.42 mg/l	
10	10	a & d & e & g & m	
11	10	a & c & g	
12	10	Random (a & f) Systematic (b&c&d&e&g&h)	
13	5	a	
Total ()			

Guide for question no. 12

- a. Titration repeatability (**Random**).
- b. Analyte loss during the sample preparation (**Systematic**). This is essentially a systematic effect. However, if parallel measurements are made then the losses usually differ from measurement to measurement. Thus, there is also a small random effect involved.
- c. Titrant decomposition during the analysis (**Systematic**). This is essentially a systematic effect. However, if parallel measurements are made then the decomposition is usually different from measurement to measurement. Thus, there is also a small random effect involved.
- d. Matrix effect in liquid chromatography mass spectrometry leading to partial suppression of ionization of the analyte (**Systematic**). This is essentially a systematic effect. However, if parallel measurements are made then the matrix effect can slightly differ from measurement to measurement. Thus, there is also a small random effect involved.

- e. Less than 100% purity of the standard substance used for constructing calibration graph (**Systematic**).
- f. Instrument reading repeatability (**Random**).
- g. Contamination of sample with the analyte (**Systematic**). This is essentially a systematic effect. However, if parallel measurements are made then the extent of contamination can slightly differ from measurement to measurement. Thus, there is also a small random effect involved.
- h. Increased absorbance readings in spectrophotometric analysis caused by interfering substance (**Systematic**). This is essentially a systematic effect. However, if parallel measurements are made then the extent of interference can slightly differ from measurement to measurement. Thus, there is also a small random effect involved

10. References:

1- A Beginner's Guide to Uncertainty of Measurement

By : Stephanie Bell

2- *Estimation of measurement uncertainty in chemical analysis*

By : Ivo Leito